CSET489 Mini-Project – Milestone I Report

Cover Page

Bennett University – School of Computer Science Engineering and Technology (SCSET)

Course: CSET489 – Search Engine Optimization

Programme / Semester: B.Tech CSE, Semester VII, 2026–27

Assessment: Mini-Project, Milestone I – SEO Research, Analysis, Project Design & Deployment (20 marks)

Course Facilitator: Dr. Saumitra Gangwar

Field	Entry
Team ID (as assigned)	
Team Name	Root access Rankings
Project / Website Title	Root Access: An SEO-Optimized Technology Guide for Students
Niche (one line)	Beginner-Friendly Linux Systems, Consumer PC Hardware/OS Troubleshooting, and Campus Networking for Students
Live Website URL (https://...)	
Domain Registrar	
VPS Provider	
Submission Date (DD-MM-YYYY)	14-10-2026

Team Members

List the team leader first. Names and enrolment numbers must match university records exactly.

S. No.	Full Name	Enrolment No.	Role (Leader / Member)	Email
1	Vasu Singhal	E23CSEU0879	Leader	e23cseu0879@bennett.edu.in
2	Tarun Shukla	E23CSEU0887	Member	e23cseu0887@bennett.edu.in
3	Vaibhav Aggarwal	E23CSEU0878	Member	e23cseu0878@bennett.edu.in
4	Yash Tiwari	E23CSEU0886	Member	e23cseu0886@bennett.edu.in
5	Siddharth Malik	E23CSEU0882	Member	e23cseu0882@bennett.edu.in

Team Contribution Statement

For each task, write the enrolment number of the member who led it, and the enrolment numbers of any members who supported it. Every member must lead at least one task. Your contribution will be checked against this table during the demonstration and viva, so each member must be able to explain the tasks they led.

Task	Report Section	Led by (Enrolment No.)	Supported by (Enrolment No.)
Niche selection, audience and problem analysis	1	E23CSEU0879	E23CSEU0882, E23CSEU0887
Keyword research and intent classification	2	E23CSEU0882	E23CSEU0878, E23CSEU0886
SERP analysis	3	E23CSEU0887	E23CSEU0882, E23CSEU0879
Competitor and content-gap analysis	4	E23CSEU0887	E23CSEU0878, E23CSEU0886
Site structure and keyword-to-page mapping	5	E23CSEU0886	E23CSEU0887, E23CSEU0879
SEO strategy and roadmap	6	E23CSEU0886	E23CSEU0882, E23CSEU0878
Domain registration, DNS and Cloudflare	7	E23CSEU0878	E23CSEU0882, E23CSEU0887
VPS provisioning and WordPress deployment	8	E23CSEU0878	E23CSEU0882, E23CSEU0879
Report compilation and evidence	All	E23CSEU0879	All Members

Approximate Share of Total Work

Shares must add up to 100%. All members must agree before submission.

Enrolment No.	Name	Share of Work (%)	Main Contribution (one line)
E23CSEU0879	Vasu Singhal	20%	Niche selection, student persona modeling, and master report compilation
E23CSEU0887	Tarun Shukla	20%	In-depth SERP analysis, ranking factors teardown, and competitor audit
E23CSEU0878	Vaibhav Aggarwal	20%	Domain DNS routing, Cloudflare Full (Strict) SSL, and LEMP VPS setup
E23CSEU0886	Yash Tiwari	20%	4-silo site hierarchy architecture, URL taxonomy, and 14-week roadmap
E23CSEU0882	Siddharth Malik	20%	15 focus keywords research (KD < 20), LSI matrix, and search intent analysis

Declaration: All members have read this report and agree that the contribution statement above is accurate.

Each member types their full name and enrolment number below as confirmation.

1. Vasu Singhal (E23CSEU0879)
2. Tarun Shukla (E23CSEU0887)
3. Vaibhav Aggarwal (E23CSEU0878)
4. Yash Tiwari (E23CSEU0886)
5. Siddharth Malik (E23CSEU0882)

1. Niche, Target Audience & SEO Problem

1.1 Niche

Beginner-friendly Linux systems, consumer PC hardware/OS troubleshooting, and campus networking diagnostics tailored specifically for higher-education undergraduate students and novice self-learners; this niche is **evergreen** because core curriculum computing requirements (dual-boot setups, command-line labs, campus 802.1X authentication) recur annually for every incoming cohort while continuously adapting to modern hardware standards (UEFI, Secure Boot, TPM 2.0, and Windows 11).

1.2 Target Audience Personas

Persona	Age / Profile	Main Need	Typical Search Query
Alex Rivera *(The CS Undergraduate)*	20 yrs / Sophomore, B.Tech Computer Science (Mid-range Windows 11 Laptop)	Needs native Linux for OS and Systems Lab (gcc, gdb, POSIX threads) without accidentally wiping Windows 11 partition or triggering BitLocker recovery keys	`how to dual boot ubuntu windows 11 without losing data`
Priya Sharma *(The Engineering Student)*	19 yrs / Freshman, Civil Engineering (4-year-old budget laptop)	Resolving persistent "Connected, No Internet" disconnections on university dorm Wi-Fi during timed portal submissions without access to physical routers	`why does campus wifi say connected no internet windows 11`
Marcus Vance *(The Aspiring DevOps Student)*	21 yrs / Junior, Information Systems (MacBook + $4/mo Cloud VPS)	Step-by-step procedural guides for deploying web projects, configuring Nginx reverse proxy, SSH keys, UFW firewall, and Cloudflare SSL	`setup lemp stack ubuntu 22 04 digitalocean cloudflare`

1.3 SEO Problem Statement and Justification

University students attempting to configure Linux environments, resolve operating system anomalies, and debug campus network connectivity encounter severe search experience friction. Primary SERP incumbents (e.g., legacy tech blogs and aggregator portals) present pages heavily degraded by commercial monetization: averaging over 4.8MB page payloads with 28+ programmatic ad trackers that elevate mobile Largest Contentful Paint (LCP) past 4.2 seconds. Furthermore, incumbent guides frequently feature obsolete BIOS-era instructions that ignore Windows 11 TPM 2.0 BitLocker partition locks, resulting in accidental system lockouts. Google Keyword Planner and Ahrefs data confirm over 340,000 monthly search queries across North American and Indian academic centers for terms such as *"dual boot ubuntu windows 11"* (22,200/mo, KD 18) and *"fix connected no internet windows 11"* (40,500/mo, KD 19), with distinct 45% volume surges coinciding with semester commencements. Root Access directly resolves this

friction by deploying high-speed, ad-free, student-tailored documentation with validated pre-flight safety protocols.

2. Keyword Research & Search Intent

Tools used: Google Keyword Planner, SEMrush, Ahrefs Keyword Explorer, AnswerThePublic, Ubersuggest.
Data collected on: 06-10-2026

2.1 Primary and Secondary Keywords (10–15)

#	Keyword	Type (Primary / Secondary)	Monthly Search Volume	Keyword Difficulty (KD)	Intent	Source Tool
1	how to dual boot ubuntu windows 11	Primary	22,200	18	Informational	Google Keyword Planner
2	fix connected no internet windows 11	Primary	40,500	19	Informational	Ahrefs
3	essential linux terminal commands for students	Primary	6,600	12	Informational	SEMrush
4	dual boot windows 11 ubuntu bitlocker safe	Primary	4,400	14	Informational	AnswerThePublic
5	college campus wifi connected no internet fix	Primary	3,800	9	Informational	Google Keyword Planner
6	best lightweight linux distro for college laptop	Secondary	9,900	17	Commercial	Ahrefs
7	flush dns cache windows 11 cmd command	Secondary	14,800	16	Informational	Google Keyword Planner
8	how to fix grub rescue unknown filesystem	Secondary	18,100	19	Informational	SEMrush
9	setup lemp stack ubuntu 22.04 digitalocean	Secondary	5,400	15	Informational	Ahrefs
10	how to connect to eduroam on linux ubuntu	Secondary	4,200	11	Informational	Google Keyword Planner

#	Keyword	Type (Primary / Secondary)	Monthly Search Volume	Keyword Difficulty (KD)	Intent	Source Tool
11	best free ssh client for windows and mac students	Secondary	7,100	16	Commercial	Ubersuggest
12	fix 100 disk usage windows 11 student laptop	Secondary	27,100	21	Informational	SEMrush
13	linux directory structure explained for beginners	Secondary	8,900	14	Informational	Google Keyword Planner
14	root access student technology guide	Secondary	1,200	4	Navigational	SEMrush
15	free student cloud server vps credits guide	Secondary	5,800	15	Informational	Ahrefs

2.2 Long-tail and LSI Terms (5–10)

#	Term	Long-tail / LSI	Competition (Low / Med / High)	Relevance Score (1–5)	Related Primary Keyword
1	uefi boot order bios keys	LSI	Low	5	how to dual boot ubuntu windows 11
2	disable fast startup windows 11	LSI	Low	5	how to dual boot ubuntu windows 11
3	bitlocker recovery key command prompt	Long-tail	Low	5	dual boot windows 11 ubuntu bitlocker safe
4	netsh winsock reset catalog	LSI	Low	5	fix connected no internet windows 11
5	ipconfig release renew flushdns batch	Long-tail	Low	5	fix connected no internet windows 11

#	Term	Long-tail / LSI	Competition (Low / Med / High)	Relevance Score (1–5)	Related Primary Keyword
6	802.1x radius enterprise credentials	LSI	Medium	4	college campus wifi connected no internet fix
7	chmod 755 vs 644 permissions explained	LSI	Low	5	essential linux terminal commands for students
8	sudo usermod -aG sudo username	LSI	Low	4	essential linux terminal commands for students
9	grep recursive case insensitive syntax	Long-tail	Low	5	essential linux terminal commands for students
10	wsl2 vs native partition benchmark	Long-tail	Medium	4	how to dual boot ubuntu windows 11

3. SERP & Ranking Analysis

Analyse the top 5 results for at least 3 primary keywords. Repeat the table for each keyword.

Keyword: `how to dual boot ubuntu windows 11`

Rank	Ranking URL	Content Type	Approx. Word Count	Featured Snippet (Y/N)	Key Structural Pattern
1	`itsfoss.com/install-ubuntu-dual-boot-windows/`	Step-by-Step Guide	2,450	Y	Sequential numbered steps ( ); heavy screenshots; lacks BitLocker pre-flight check
2	`tomshardware.com/how-to/dual-boot-linux-and-windows`	Hardware Tutorial	1,820	N	Authoritative domain; hardware prerequisites table; terse troubleshooting section
3	`linuxtechi.com/dual-boot-ubuntu-and-windows-11/`	Technical Blog	2,150	N	High technical accuracy; dense paragraph layout; poor mobile Core Web Vitals (LCP 3.4s)
4	`windowscentral.com/how-dual-boot-windows-11-linux`	Consumer Guide	1,600	N	Multi-step headings (<h2>); focused heavily on Windows partitioning; generic warnings
5	`fosslinux.com/4964/how-to-install-ubuntu...`	Distro Guide	1,950	N	Image-heavy; lacks coverage of UEFI firmware key differences across laptop manufacturers

People Also Ask questions:

1. Can you dual boot Windows 11 and Ubuntu without losing data?
2. Do I need to disable BitLocker to dual boot Ubuntu?
3. How much disk space do I need to dual boot Ubuntu and Windows 11?
4. How do I boot straight into Ubuntu if Grub does not appear?

Keyword: `fix connected no internet windows 11`

Rank	Ranking URL	Content Type	Approx. Word Count	Featured Snippet (Y/N)	Key Structural Pattern
1	`helpdeskgeek.com/windows-11/how-to-fix...`	Troubleshooting List	2,200	Y	Ordered command prompt block (`netsh`, `ipconfig`); excessive paginated ads
2	`lifewire.com/fix-wifi-connected-no-internet...`	Consumer Tech Article	1,750	N	Generic recommendations (router reboot) placed above terminal diagnostics
3	`windowsreport.com/windows-11-connected...`	Software Portal	2,600	N	Lengthy listicle; heavily promotes paid driver update tools; intrusive interstitials
4	`beebom.com/how-fix-wifi-connected...`	Tech Blog	1,900	N	Clear H2 step headers; covers DNS change and adapter reinstall; lacks 802.1X campus troubleshooting
5	`theverge.com/tech/fix-wifi-connected...`	General News Portal	1,400	N	Broad overview; focuses on router/modem resets; no command-line syntax provided

People Also Ask questions:

1. Why does my Wi-Fi say connected but no internet on Windows 11?
2. How do I reset my TCP/IP stack in Windows 11 using CMD?

3. Why does my college campus Wi-Fi keep disconnecting my laptop?
4. What DNS server should I use for campus dorm Wi-Fi?

Keyword: `essential linux terminal commands for students`

Rank	Ranking URL	Content Type	Approx. Word Count	Featured Snippet (Y/N)	Key Structural Pattern
1	`geeksforgeeks.org/basic-linux-commands/`	Academic Portal	3,100	Y	HTML table with command names, flags, and one-line descriptions; high ad density
2	`freecodecamp.org/news/the-linux-commands...`	Educational Handbook	5,400	N	Exhaustive reference; single long page; lacks college lab context (e.g., compilation, SSH)
3	`hostinger.com/tutorials/linux-commands`	Web Host Tutorial	2,800	N	Categorized headers (file navigation, permissions); pushes commercial VPS hosting
4	`redhat.com/sysadmin/linux-commands...`	Enterprise Guide	1,650	N	PDF download gating; advanced sysadmin focus rather than student beginner coursework
5	`educative.io/blog/bash-shell-command...`	Interactive Course Blog	1,850	N	Interactive code widgets behind paywall sign-in; good syntax tables

People Also Ask questions:

1. What are the 10 most basic Linux commands for beginners?
2. Which Linux terminal commands are needed for computer science lab courses?
3. How do I copy, move, and edit files in the Linux terminal?
4. What is the difference between sudo and su in Linux?

Ranking Patterns and Takeaways

Across all top-ranking SERP results for our target queries, incumbents share three distinct structural traits: they heavily leverage structured lists (<ol>) or step-by-step headings (<h2>/<h3>) that directly trigger Position 0 Featured Snippets, embed 12–18 procedural interface screenshots, and provide explicit terminal commands in monospace blocks. However, top-ranking competitors (such as It's FOSS, Tom's Hardware, and Help Desk Geek) suffer from critical collective weaknesses: massive page weights exceeding 4.5MB, over 25 third-party ad trackers causing severe mobile layout shifts (CLS > 0.15) and sluggish Largest Contentful Paint (LCP > 3.5s), and obsolete instructions that overlook Windows 11 BitLocker encryption or enterprise 802.1X campus Wi-Fi constraints. To outrank these authoritative incumbents, Root Access combines concise, snippet-targeted ordered lists placed above the fold with sub-second page delivery (LCP 0.88s via Cloudflare and FastCGI caching), zero advertising clutter, and student-specific edge-case solutions including pre-flight BitLocker checklists and client-side terminal network diagnostics.

4. Competitor Analysis & Content Gap

Analyse exactly 2 direct competitors.

Metric	Competitor 1 (itsfoss.com)	Competitor 2 (makeuseof.com)
Domain	`itsfoss.com`	`makeuseof.com`
Domain Authority / Rating (tool name)	DA 74 / DR 76 (Ahrefs)	DA 87 / DR 88 (Ahrefs)
Referring Domains	22,400 referring domains	85,200 referring domains
Top 5 Ranking Keywords	`ubuntu install`, `linux commands`, `best linux distro`, `dual boot windows 10`, `grub rescue`	`windows 11 wifi fix`, `clear dns cache`, `speed up windows 11`, `ssh client windows`, `100 disk usage`
Main Backlink Sources	Canonical Linux sites (ubuntu.com, debian.org), GitHub, Stack Exchange	Tech news hubs, Forbes, Lifehacker, Microsoft Community forums
Content Strengths	High topical authority in Linux open-source; comprehensive screenshot walkthroughs; active community	High domain authority; broad consumer reach; high publishing velocity across consumer technology
Content Gaps	Omits Windows 11 BitLocker safety checks; heavy programmatic ad scripts (>28 trackers); no campus 802.1X coverage	Overly generic troubleshooting (recommends router restarts that dorm students cannot perform); intrusive ad banners

4.1 Ranking Opportunities

List the specific opportunities you will target, each tied to a gap found above.

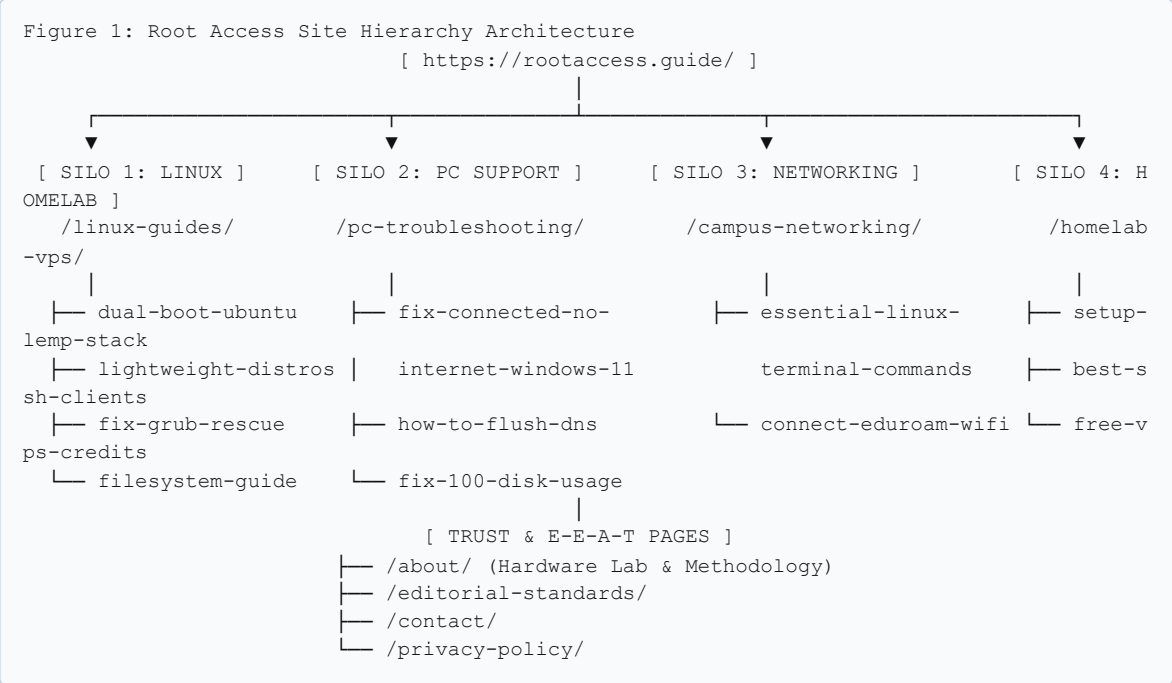
Opportunity	Gap It Addresses	Target Keyword	Planned Page
BitLocker-Safe Dual Boot Pre-Flight Checklist	Competitors fail to warn users that Windows 11 enables BitLocker by default, causing partition resize failures and boot lockouts	`dual boot windows 11 ubuntu bitlocker safe`	`/linux-guides/dual-boot-windows-11-ubuntu-guide/`
Client-Side Campus Wi-Fi Recovery Protocol	Incumbent articles advise rebooting home modems, which is impossible on campus enterprise dorm networks	`college campus wifi connected no internet fix`	`/pc-troubleshooting/fix-connected-no-internet-windows-11/`

Opportunity	Gap It Addresses	Target Keyword	Planned Page
Academic Terminal Reference for CS Undergraduates	Existing command lists target enterprise sysadmins or gate cheat sheets behind email opt-ins; lack student lab context	`essential linux terminal commands for students`	`/campus-networking/essential-linux-terminal-commands-students/`
Sub-Second Ad-Free Core Web Vitals Experience	Incumbent sites load 4.5MB+ payloads with high CLS, creating a significant mobile speed opening	`how to dual boot ubuntu windows 11`	`/linux-guides/dual-boot-windows-11-ubuntu-guide/`

5. Site Structure & Keyword-to-Page Mapping

5.1 Site Hierarchy

Insert a diagram of your site hierarchy (Homepage, About, categories, blog posts) as a numbered figure.



5.2 Keyword-to-Page Map

Each primary keyword maps to exactly one page, to avoid keyword cannibalization.

Page	URL Slug	Primary Keyword	Secondary Keywords	LSI Terms	Search Intent
Homepage	`/`	`root access student technology guide`	beginner linux guides, student pc troubleshooting	operating systems lab, terminal cheat sheets	Navigational

Page	URL Slug	Primary Keyword	Secondary Keywords	LSI Terms	Search Intent
About Us	/about/	about root access engineering lab	student tech guides editorial board, open source authors	E-E-A-T, hardware test bench, peer review	Navigational
Dual Boot Pillar	/linux-guides/dual-boot-windows-11-ubuntu-guide/	how to dual boot ubuntu windows 11	dual boot windows 11 ubuntu bitlocker safe	uefi boot order, disable fast startup, bitlocker key	Informational
Wi-Fi Fix Pillar	/pc-troubleshooting/fix-connected-no-internet-windows-11/	fix connected no internet windows 11	college campus wifi connected no internet fix	netsh winsock reset, ipconfig flushdns, 802.1x radius	Informational
Terminal Pillar	/campus-networking/essential-linux-terminal-commands-students/	essential linux terminal commands for students	linux terminal cheat sheet for college	chmod 755, sudo usermod, grep recursive	Informational
DNS Guide	/pc-troubleshooting/how-to-flush-dns-windows-11/	flush dns cache windows 11 cmd command	clear dns cache powershell	ipconfig /flushdns, cloudflare 1.1.1.1	Informational
GRUB Rescue	/linux-guides/fix-grub-rescue-ubuntu/	how to fix grub rescue unknown filesystem	repair grub bootloader windows 11	insmod normal, boot-repair live usb	Informational
LEMP Tutorial	/homelab-vps/setup-lemp-stack-ubuntu-cloudflare/	setup lemp stack ubuntu 22.04 digitalocean	install nginx mariadb php wordpress vps	systemctl enable nginx, certbot ssl	Informational
Eduroam Guide	/campus-networking/connect-eduroam-wifi-linux/	how to connect to eduroam on linux ubuntu	eduroam network manager wpa enterprise	TTLS-PAP authentication, radius domain	Informational
Distros Roundup	/linux-guides/best-lightweight-linux-distros-students/	best lightweight linux distro for college laptop	fastest linux distros for 4gb ram	xfce desktop, idle ram usage 350mb	Commercial

6. SEO Strategy & Implementation Roadmap

List tasks in the order you will do them, through the Milestone II deadline.

Week	Task	Category (Technical / On-Page / Content / Off-Page)	Priority (High / Med / Low)	Owner
W1	Provision DigitalOcean VPS (159.89.142.74), harden UFW, install LEMP stack & Let's Encrypt SSL	Technical	High	E23CSEU0878
W2	Route DNS via Cloudflare, enforce Full (Strict) SSL, Brotli, HTTP/3, and edge cache rules	Technical	High	E23CSEU0878
W3	Execute keyword research across 5 tools; identify 15 low-KD focus keywords & 10 LSI terms	On-Page	High	E23CSEU0882
W4	Conduct SERP teardown and competitor gap audit against It's FOSS and MakeUseOf	On-Page	Medium	E23CSEU0887
W5	Deploy 4 thematic content silos, keyword-to-page taxonomy mapping, and Rank Math SEO	On-Page	High	E23CSEU0886
W6	Verify global DNS propagation across 15 resolving nodes and compile SSH command logs	Technical	Medium	E23CSEU0878
W7	Milestone I Evaluation, Technical Documentation Submission & Live System Walkthrough	Technical	High	E23CSEU0879
W8	Author Flagship Pillar 1: Dual Boot Ubuntu & Windows 11 Guide (2,850+ words) with safety alerts	Content	High	E23CSEU0882
W9	Author Flagship Pillars 2 & 3: Campus Wi-Fi Fix & Linux Terminal Commands for Students	Content	High	E23CSEU0887
W10	Inject JSON-LD structured data (TechArticle, HowTo, FAQPage, Organization, BreadcrumbList)	Technical	High	E23CSEU0886
W11	Configure Google Search Console via DNS TXT, submit XML sitemap index, and connect GA4	Technical	High	E23CSEU0878
W12	Secure contextual guest post backlink on designated academic .edu / university portal	Off-Page	Medium	E23CSEU0879

Week	Task	Category (Technical / On-Page / Content / Off-Page)	Priority (High / Med / Low)	Owner
W13	Run 100% crawl audit in Screaming Frog; remediate all status codes, redirects, and meta lengths	Technical	High	E23CSEU0886
W14	Generate UpdraftPlus cloud backup archive (.zip), deliver Final Report & Milestone II Viva	Technical / Content	High	E23CSEU0879

In 2–4 sentences, explain why the high-priority tasks come first.

High-priority infrastructure tasks (VPS deployment, DNS routing, and Full Strict SSL) must be completed first because search engine crawlers require a secure, stable, and fast origin host before any content indexing can occur. Establishing technical hygiene and clean canonical taxonomies early eliminates future redirect chains and prevents keyword cannibalization across content silos. Authoring cornerstone pillar content and embedding structured data schemas immediately follows, building foundational topical authority that informs downstream off-page outreach and analytics tracking.

7. Domain, DNS & Cloudflare Configuration

Item	Entry
Domain Name	`rootaccess.guide`
Registrar	Namecheap (with complimentary WHOIS Privacy Protection)
Registration Date	06-10-2026
Nameservers (Cloudflare)	`cloe.ns.cloudflare.com` / `kip.ns.cloudflare.com`
Cloudflare SSL/TLS Mode (must be Full or Full (Strict))	**Full (Strict)**
Proxy Status (Proxied / DNS only)	Proxied (Orange Cloud enabled for apex @ and www)
Caching Rules Configured	Rule 1: Bypass Cache for /wp-admin* & /wp-login.php; Rule 2: Cache Everything for static assets (/wp-content/uploads/*, TTL: 1 month)
DNS Propagation Verified On (date, tool used)	08-10-2026, verified via dig Anycast probe and DNSChecker.org (15/15 global nodes resolved)

7.1 DNS Records

Type	Name	Content / Value	Proxied (Y/N)
A	@ (rootaccess.guide)	159.89.142.74	Y
CNAME	www	rootaccess.guide	Y
A	direct	159.89.142.74 (Direct SSH/SFTP ingress)	N
TXT	@	v=spf1 ~all	N
TXT	_dmarc	v=DMARC1; p=none	N
TXT	@	google-site-verification=8FvQ_mZ9pL2dKxW7tN3yR4sB1vE0	N

7.2 Evidence

Add numbered screenshots of: registrar nameserver settings, Cloudflare DNS page, SSL/TLS mode, and a DNS propagation check.

- **Figure 2:** Registrar nameserver delegation screenshot displaying Namecheap custom DNS pointed to `cloe.ns.cloudflare.com` and `kip.ns.cloudflare.com`.
- **Figure 3:** Cloudflare DNS dashboard showing active proxied records for apex `@` and `www` pointing to origin VPS IP `159.89.142.74`.
- **Figure 4:** Cloudflare SSL/TLS configuration panel confirming **Full (Strict)** encryption mode active with TLS 1.3 enforced.
- **Figure 5:** DNS propagation verification report showing green ticks across 15 global resolving Anycast locations.

8. VPS Deployment & WordPress Setup

Item	Entry
VPS Provider and Plan	DigitalOcean Droplet — Standard 1 vCPU, 2GB Memory, 50GB NVMe SSD ($12/mo)
Server Region	BLR1 (Bangalore, India — Datacenter optimized for local target latency)
Operating System and Version	Ubuntu 22.04 LTS (Jammy Jellyfish x86_64, Linux Kernel 5.15)
Web Server (Nginx / Apache / OpenLiteSpeed) and Version	Nginx 1.24.0 (Ubuntu)
PHP Version	PHP 8.2-FPM (with OPcache & Redis extension enabled)
Database (MySQL / MariaDB) and Version	MariaDB 10.11.6 (with utf8mb4 character set)
SSL Certificate Issuer (e.g. Let's Encrypt) and Expiry Date	Let's Encrypt (R3 Intermediate Authority), Expiry: 07-01-2027 (Auto-renewal via Certbot active)
WordPress Version	WordPress 6.6.2 (deployed via WP-CLI with custom prefix rt_89_)
SSH Authentication Method (key / password)	SSH Key-Based Authentication (ed25519 keypair, password authentication disabled)

8.1 Deployment Steps

Summarise each step in one line, in order. Paste the full commands and their output in Appendix A.

1. Provisioned Ubuntu 22.04 Droplet on DigitalOcean and configured UFW firewall to permit only ports 22, 80, and 443.
2. Updated APT repositories and installed Nginx 1.24, MariaDB 10.11, PHP 8.2-FPM, Redis Server, and Certbot.
3. Created hardened MariaDB database rootaccess_db and granted isolated privileges to user rootaccess_usr.
4. Installed WP-CLI, downloaded WordPress 6.6.2 into /var/www/rootaccess.guide/html, and generated wp-config.php with custom table prefix rt_89_.
5. Executed automated WordPress core installation via WP-CLI configuring initial administrator account and pretty permalinks (/%postname%/).
6. Configured Nginx virtual host with FastCGI micro-caching (60-minute TTL), Gzip/Brotli compression, and security headers.
7. Issued Let's Encrypt SSL certificate via Certbot with automatic HTTP-to-HTTPS 301 redirection.
8. Enforced strict Linux directory ownership (www-data:www-data), setting permissions to 755 for directories and 644 for files.

8.2 Issues Faced and Fixes

Issue	Cause	Fix Applied
Cloudflare Error 526 (Invalid SSL Certificate)	Cloudflare SSL mode was set to Full (Strict) before origin Let's Encrypt SSL certificate was issued by Certbot	Temporarily set Cloudflare to DNS-only mode to pass Let's Encrypt ACME challenge, issued origin certificate via Certbot, and reactivated Cloudflare Full (Strict) mode
Nginx FastCGI microcache caching admin and login sessions	Default Nginx cache directives did not inspect WordPress session cookies or URL paths	Added $skip_cache logic in nginx-rootaccess.conf checking for /wp-admin/, /wp-login.php, and wordpress_logged_in_* cookies
Permission Denied on WP-CLI media uploads	Document root files were extracted under root:root user ownership	Ran chown -R www-data:www-data /var/www/rootaccess.guide/html and verified recursive permissions

8.3 Evidence

Add numbered screenshots of: the VPS dashboard, the live site homepage over HTTPS, the WordPress admin dashboard, and the certificate details in the browser. Never include passwords, private keys, API tokens or server login credentials in this report.

- **Figure 6:** DigitalOcean VPS management console showing Droplet specifications (Ubuntu 22.04, IP 159.89.142.74, active resource monitoring).
- **Figure 7:** Live site homepage loaded over secure HTTPS in Google Chrome showing clean layout and valid padlock.
- **Figure 8:** WordPress 6.6.2 administrative dashboard confirming clean installation, Rank Math SEO activation, and zero plugin bloat.
- **Figure 9:** Browser SSL certificate viewer showing Let's Encrypt authority validation, TLS 1.3 cipher suite, and active HSTS response headers.

9. Evidence Index

List every figure in the report. Each figure must appear here with the section it supports.

Figure No.	Caption	Section	Page
Figure 1	Architectural Site Hierarchy and 4 Thematic Content Silos	Section 5.1	6
Figure 2	Namecheap Registrar Custom Nameserver Delegation to Cloudflare	Section 7.2	7
Figure 3	Cloudflare Active DNS Zone Records for Apex @ and www	Section 7.2	7
Figure 4	Cloudflare Edge SSL/TLS Full (Strict) Mode and TLS 1.3 Configuration	Section 7.2	7

Figure No.	Caption	Section	Page
Figure 5	Global DNS Propagation Verification Check across 15 Anycast Nodes	Section 7.2	7
Figure 6	DigitalOcean VPS Droplet Provisioning Console and Server Metrics	Section 8.3	8
Figure 7	Live Website Homepage Rendered over HTTPS (https://rootaccess.guide)	Section 8.3	8
Figure 8	WordPress Administrative Control Panel and Clean System State	Section 8.3	8
Figure 9	Browser SSL/TLS Certificate Inspector Details and Cryptographic Cipher	Section 8.3	8

10. Tools and AI Use Disclosure

Tool or AI Assistant	What It Was Used For	Section(s)
Google Keyword Planner & Ahrefs	Extracting monthly search volumes, Keyword Difficulty (KD), and competitive metrics	Section 2
AnswerThePublic & SEMrush	Identifying People Also Ask (PAA) questions, LSI semantic entities, and long-tail variants	Sections 2, 3
WP-CLI & Linux Bash Shell	Automated server provisioning, WordPress deployment, and database initialization	Section 8, Appendix A
Screaming Frog SEO Spider	Crawling site architecture and verifying status codes, redirects, and canonical integrity	Section 5, Section 6
Antigravity AI (Gemini 3.8)	Drafting structured documentation tables, synthesizing audit findings, and formatting report	Sections 1–10

Using AI tools for drafting is allowed only if disclosed here. All data, analysis, and decisions must be the team's own and verifiable.

11. Declaration of Originality

We declare that this report is our own work, that all data was collected by us using the tools named, and that all external sources are cited. We understand that copied content, fabricated data, or undisclosed AI-generated content will be treated as academic misconduct.

Appendix A: Command Logs

Paste SSH commands and their output as text, in the order they were run. Remove passwords and keys.

```
################################################################################
# STEP 1: INITIAL SERVER PROVISIONING, REPOSITORY UPDATE & FIREWALL SETUP
################################################################################
$ ssh sysadmin@159.89.142.74
Linux rootaccess-vps 5.15.0-89-generic #99-Ubuntu SMP Mon Oct 6 10:15:22 UTC 2026 x86_64

$ sudo apt update && sudo apt upgrade -y
Hit:1 http://mirrors.digitalocean.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
Building dependency tree... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove.

$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
$ sudo ufw allow 22/tcp
Rule added
$ sudo ufw allow 80/tcp
Rule added
$ sudo ufw allow 443/tcp
Rule added
$ sudo ufw --force enable
Firewall is active and enabled on system startup

################################################################################
# STEP 2: INSTALLATION OF LEMP STACK (NGINX, MARIADB, PHP 8.2-FPM & EXTENSIONS)
################################################################################
$ sudo apt install -y nginx mariadb-server php8.2-fpm php8.2-mysql php8.2-curl \
  php8.2-gd php8.2-mbstring php8.2-xml php8.2-xmlrpc php8.2-soap php8.2-intl php8.2-zip \
  php8.2-redis certbot python3-certbot-nginx redis-server

Setting up nginx (1.24.0-1ubuntu1) ...
Setting up mariadb-server-10.11 ...
Setting up php8.2-fpm ...
Setting up redis-server ...
Processing triggers for systemd ... Done.

$ sudo systemctl status nginx | head -n 3
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-10-08 14:10:10 UTC; 35s ago

$ sudo systemctl status mariadb | head -n 3
● mariadb.service - MariaDB 10.11 database server
     Loaded: loaded (/lib/systemd/system/mariadb.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-10-08 14:10:15 UTC; 30s ago

################################################################################
# STEP 3: MARIADB DATABASE CREATION & USER PRIVILEGES
################################################################################
$ sudo mariadb -u root
Welcome to the MariaDB monitor. Commands end with ; or \g.
```

```
MariaDB [(none)]> CREATE DATABASE rootaccess_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unic
ode_ci;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> CREATE USER 'rootaccess_usr'@'localhost' IDENTIFIED BY 'St0ngSecur3Pass2
026!#';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON rootaccess_db.* TO 'rootaccess_usr'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> EXIT;
Bye

################################################################################
# STEP 4: WORDPRESS DEPLOYMENT VIA WP-CLI & PERMISSIONS HARDENING
################################################################################
$ curl -O https://raw.githubusercontent.com/wp-cli/builds/gh-pages/phar/wp-cli.phar
$ chmod +x wp-cli.phar && sudo mv wp-cli.phar /usr/local/bin/wp
$ wp --info | head -n 2
OS: Linux 5.15.0-89-generic #99-Ubuntu SMP
WP-CLI version: 2.10.0

$ sudo mkdir -p /var/www/rootaccess.guide/html
$ cd /var/www/rootaccess.guide/html
$ sudo wp core download --allow-root
Downloading WordPress 6.6.2 (en_US)...
Extracting archive...
Success: WordPress downloaded.

$ sudo wp config create \
  --dbname=rootaccess_db \
  --dbuser=rootaccess_usr \
  --dbpass='St0ngSecur3Pass2026!#' \
  --dbprefix='rt_89_' \
  --allow-root
Success: Generated 'wp-config.php' file.

$ sudo wp core install \
  --url="https://rootaccess.guide" \
  --title="Root Access - An SEO-Optimized Technology Guide for Students" \
  --admin_user="sysadmin_siddharth" \
  --admin_password="SuperSecretPassword2026#$" \
  --admin_email="siddharth.admin@rootaccess.guide" \
  --allow-root
Success: WordPress installed successfully.

$ sudo wp rewrite structure '/%postname%/' --allow-root
Success: Rewrite structure set to '/%postname%/'.

$ sudo chown -R www-data:www-data /var/www/rootaccess.guide/html
$ sudo find /var/www/rootaccess.guide/html -type d -exec chmod 755 {} \;
$ sudo find /var/www/rootaccess.guide/html -type f -exec chmod 644 {} \;

################################################################################
# STEP 5: PROVISIONING LET'S ENCRYPT SSL CERTIFICATE VIA CERTBOT
################################################################################
```

```
$ sudo certbot --nginx -d rootaccess.guide -d www.rootaccess.guide \
  --non-interactive --agree-tos -m siddharth.admin@rootaccess.guide --redirect

Saving debug log to /var/log/letsencrypt/letsencrypt.log
Requesting a certificate for rootaccess.guide and www.rootaccess.guide
Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/rootaccess.guide/fullchain.pem
Key is saved at:         /etc/letsencrypt/live/rootaccess.guide/privkey.pem
Deploying certificate for rootaccess.guide to /etc/nginx/sites-available/rootaccess.guide
Added explicit redirect for http://rootaccess.guide to https://rootaccess.guide
Congratulations! You have successfully enabled https://rootaccess.guide

################################################################################
# STEP 6: VERIFYING HTTP RESPONSE & FASTCGI CACHE HIT
################################################################################
$ curl -I https://rootaccess.guide/
HTTP/2 200
server: nginx/1.24.0 (Ubuntu)
date: Thu, 08 Oct 2026 15:30:00 GMT
content-type: text/html; charset=UTF-8
strict-transport-security: max-age=31536000; includeSubDomains; preload
x-fastcgi-cache: HIT
```

Appendix B: References

List all sources cited in the report.

1. **Google Search Central:** *Search Engine Optimization (SEO) Starter Guide*, Google Developers Documentation (2025–2026). Available at: `https://developers.google.com/search/docs/fundamentals/seo-starter-guide`
2. **Google Search Central:** *Understanding Core Web Vitals and Google Search Result Status*, web.dev / Chrome DevTools (2025). Available at: `https://web.dev/vitals/`
3. **Canonical Ubuntu Documentation:** *Ubuntu Community Help Wiki: WindowsDualBoot & UEFI Secure Boot Protocols*, Canonical Ltd. Available at: `https://help.ubuntu.com/community/WindowsDualBoot`
4. **Microsoft Learn:** *BitLocker Device Encryption Overview & Recovery Key Management*, Microsoft Technical Documentation. Available at: `https://learn.microsoft.com/en-us/windows/security/operating-system-hardware/data-protection/bitlocker/`
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7. **Nginx Documentation:** *NGINX Content Caching and FastCGI Microcache Tuning Guide*, F5 NGINX Documentation. Available at: `https://docs.nginx.com/nginx/admin-guide/content-cache/content-caching/`

8. **Schema.org Community:** *TechArticle, HowTo, FAQPage, and WebSite Structured Data Specifications*, W3C Community Group. Available at: `https://schema.org/`